

ea源码使用文档

环境准备

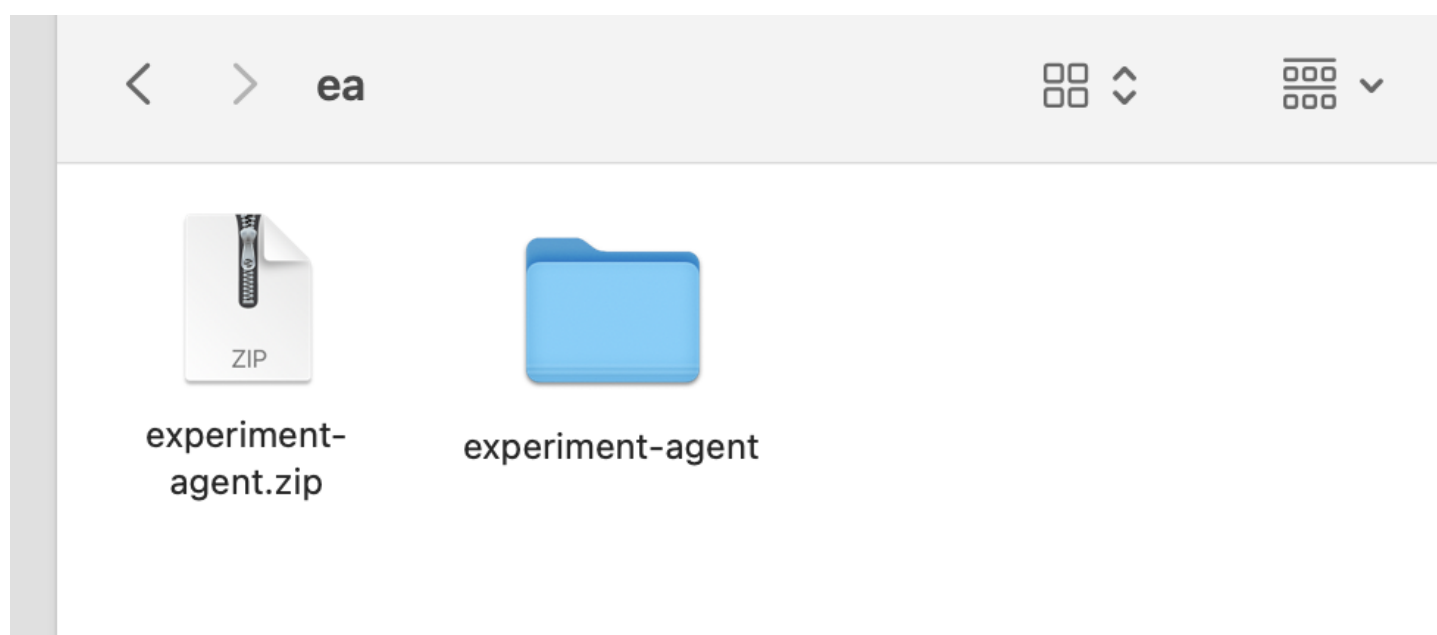
系统环境必须安装有conda与docker

源代码:

<https://github.com/hitsz-ids/AI-Course.git>

Agentic_AI_M2/ea复现/code/experiment-agent.zip

解压源代码到文件夹:

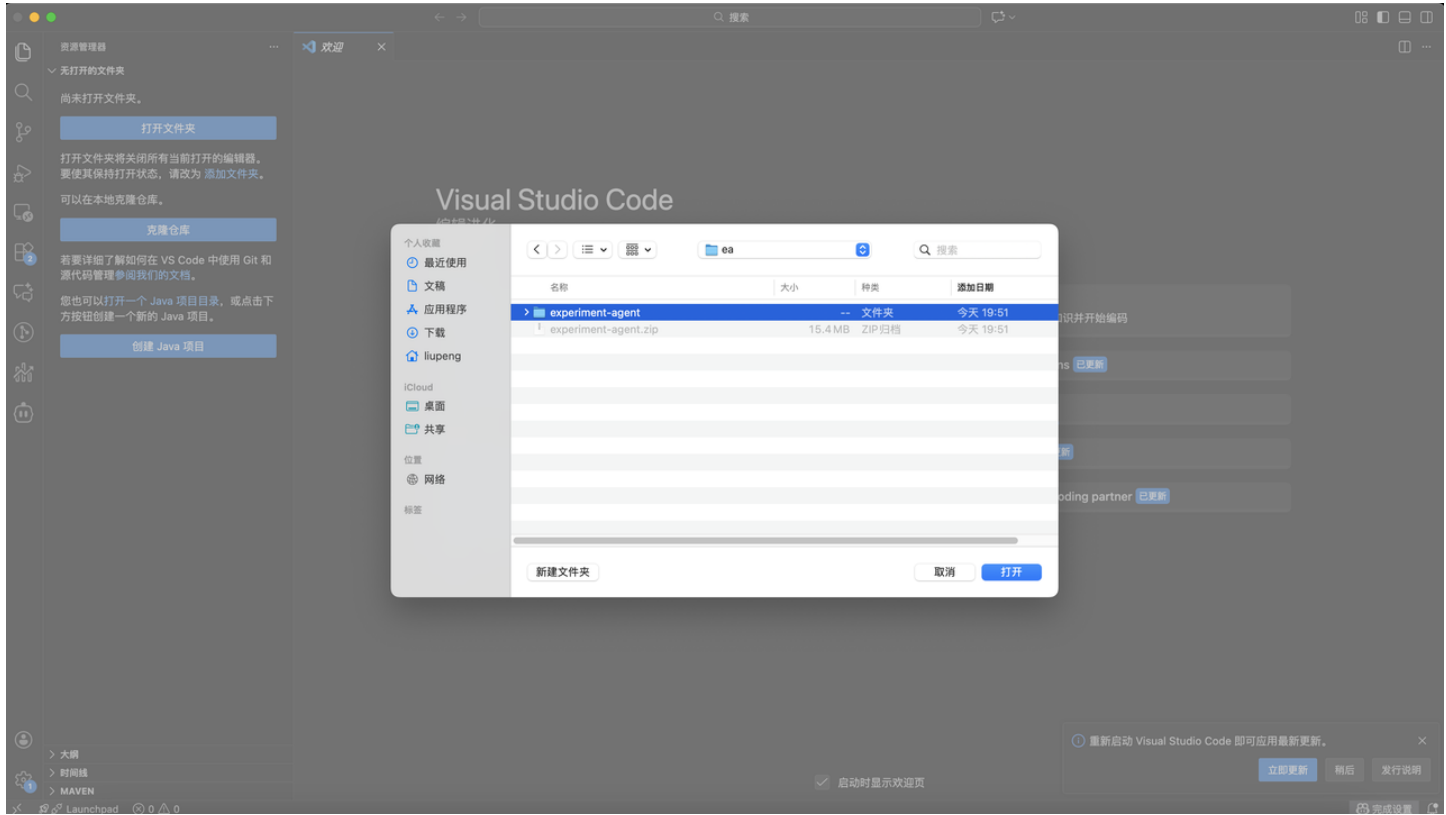


使用命令行解压:

代码块

- 1 #将压缩包放到文件夹解压
- 2 unzip experiment-agent.zip

打开VSCode选择已解压项目



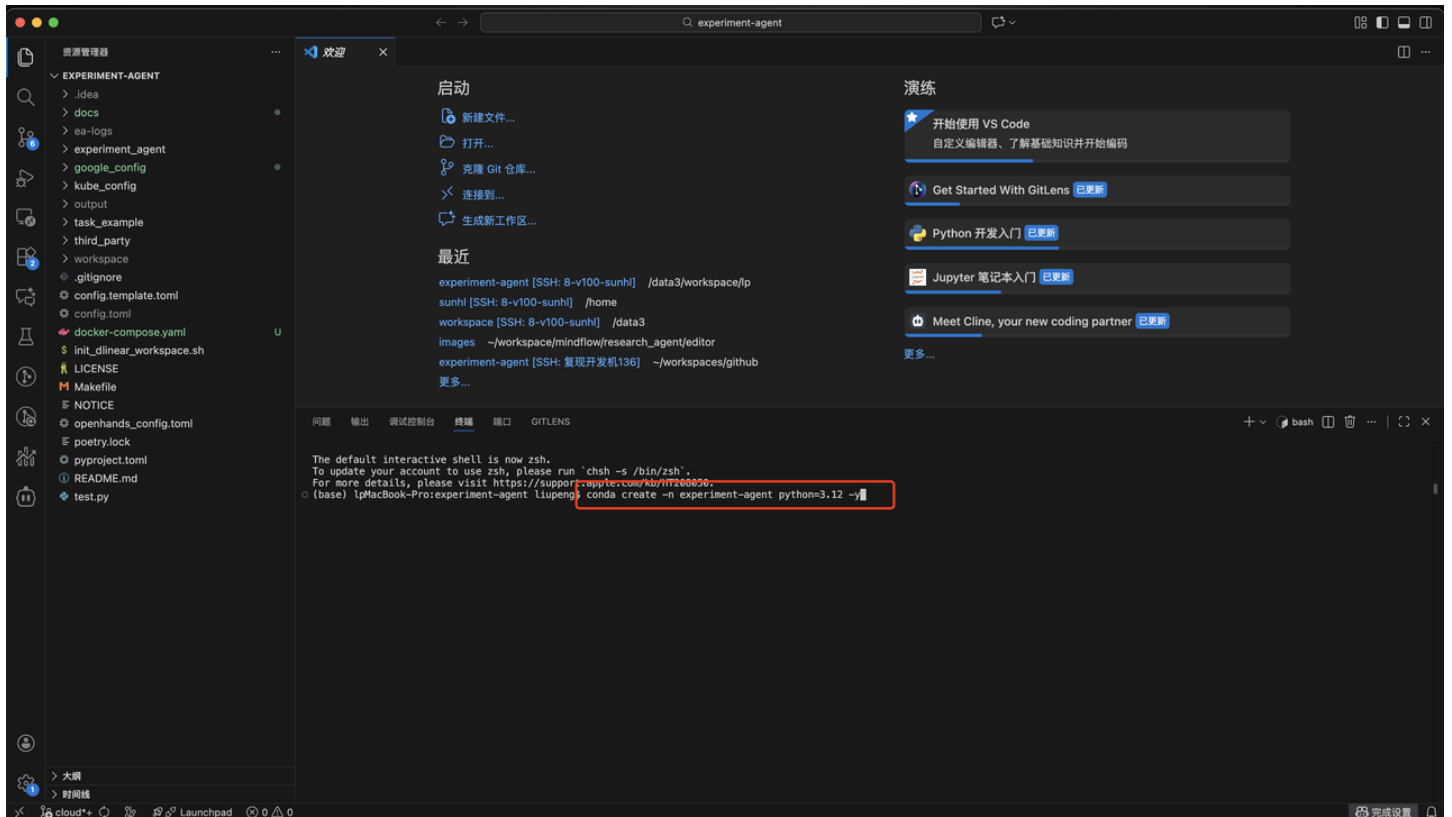
创建ea-python环境

1.新建conda环境

新建终端,并创建experiment-agent conda环境

代码块

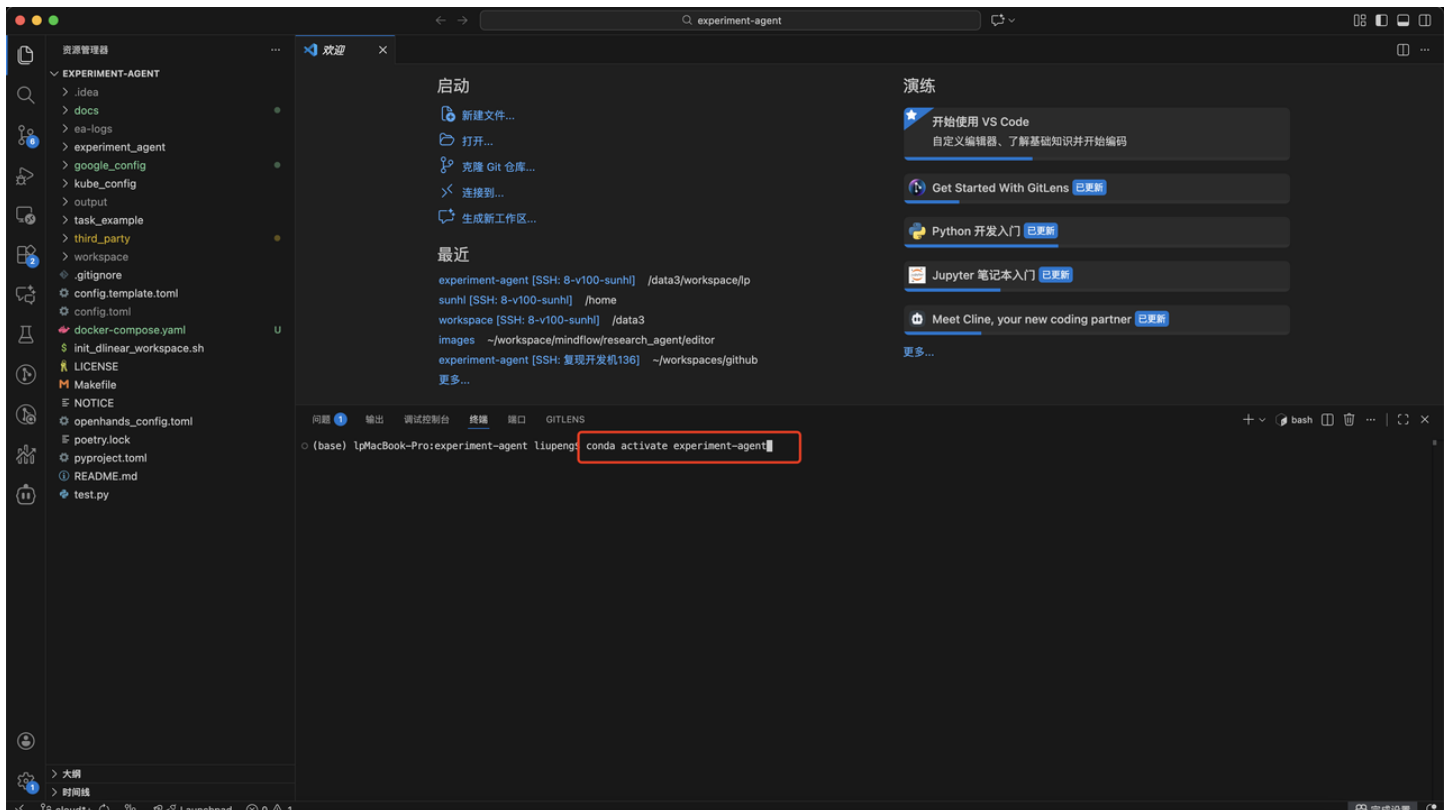
```
1      conda create -n experiment-agent python=3.12 -y
```



2. 激活conda环境

代码块

```
1 conda activate experiment-agent
```

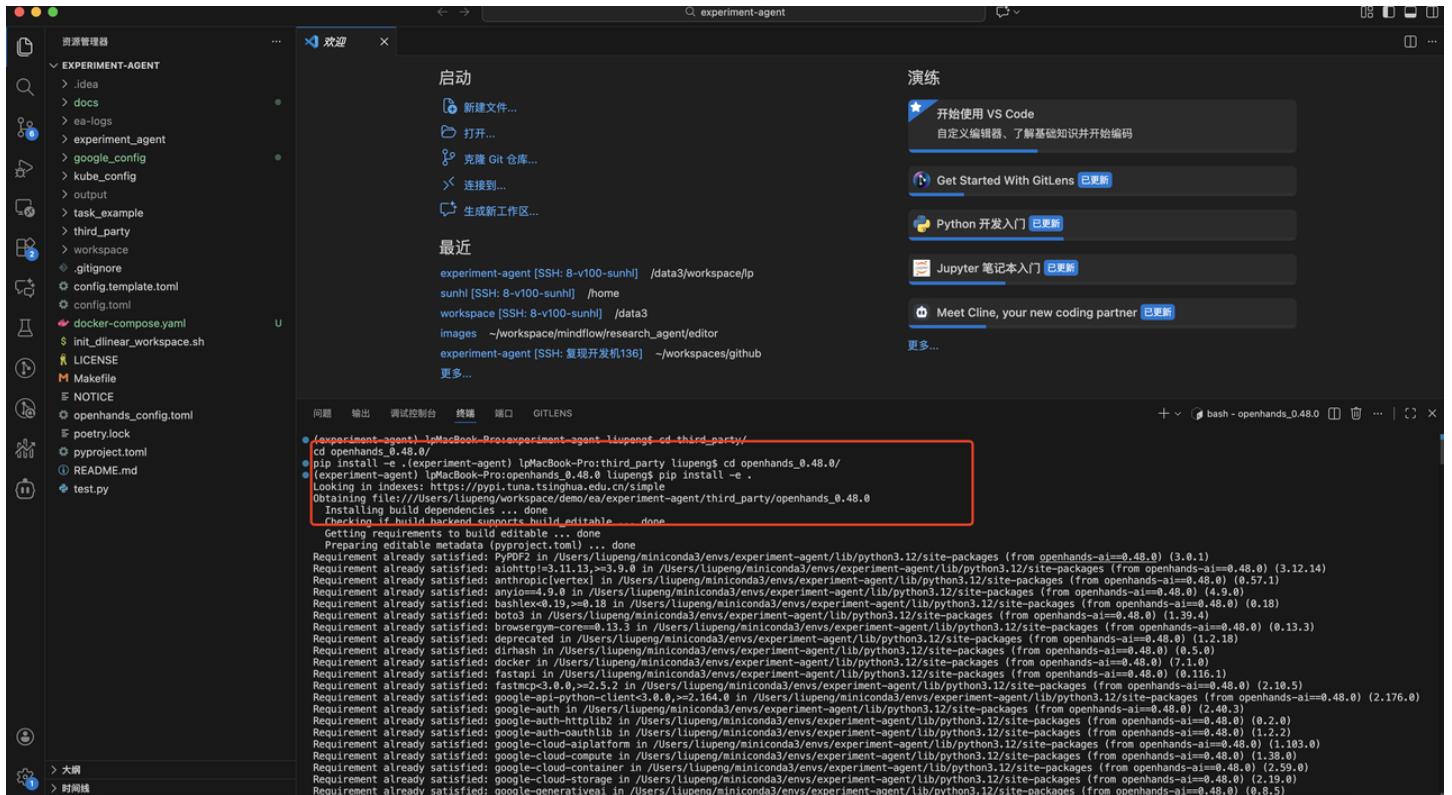


3. 依赖安装

3.1 openhands依赖安装

代码块

```
1 cd third_party/
2 cd openhands_0.48.0/
3 pip install -e .
```



完成依赖后终端指令退回到项目根目录

代码块

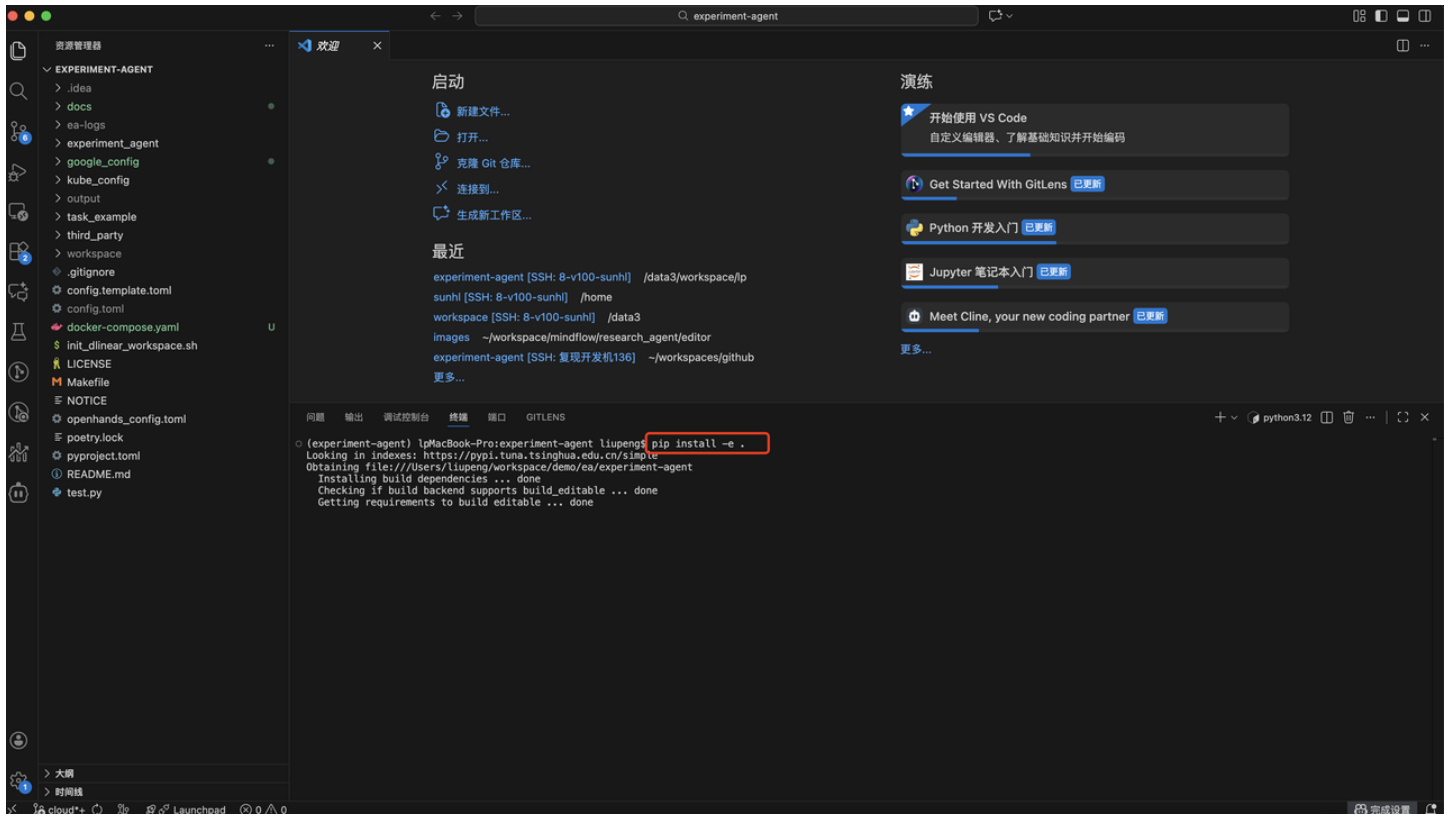
```
1 cd ../
2 cd ../
```

3.2 ea依赖安装

在ea项目根目录执行命令

代码块

```
1 pip install -e .
```



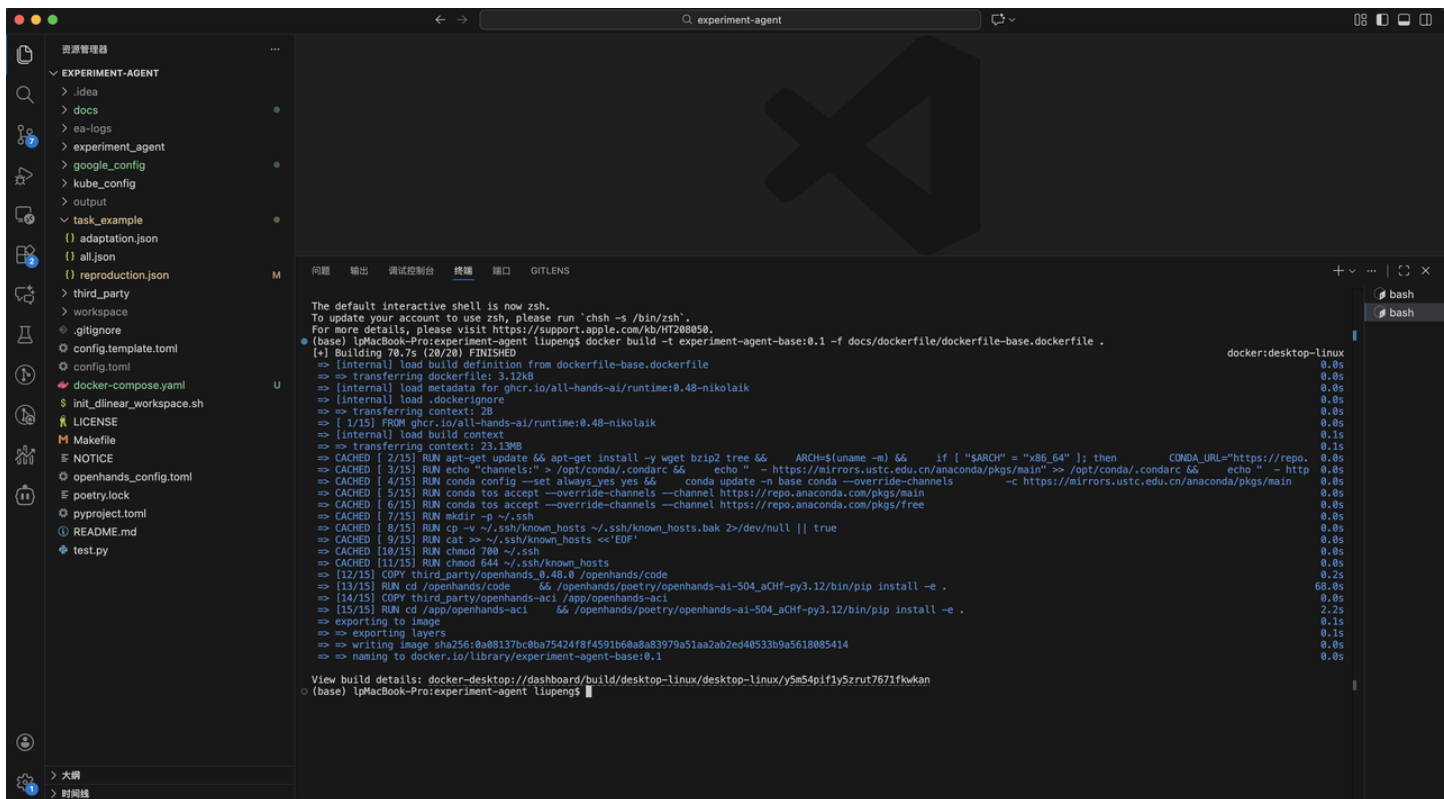
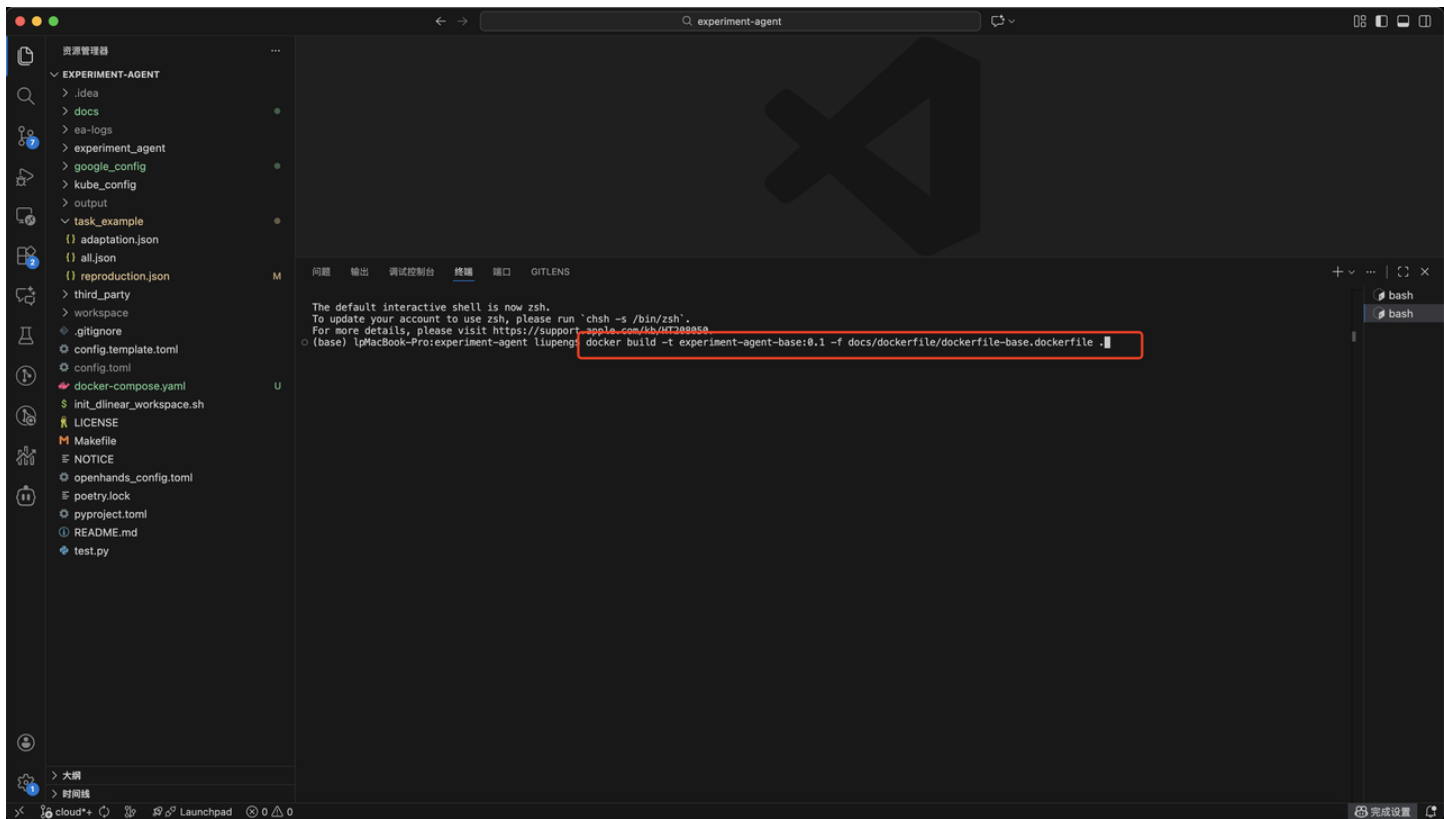
等待命令执行完毕,完成ea依赖安装

4.镜像构建

运行命令构建ea使用的基础镜像

代码块

```
1 docker build -t experiment-agent-base:0.1 -f docs/dockerfile/dockerfile-  
base.dockerfile .
```



复现TimeXer项目

检查配置

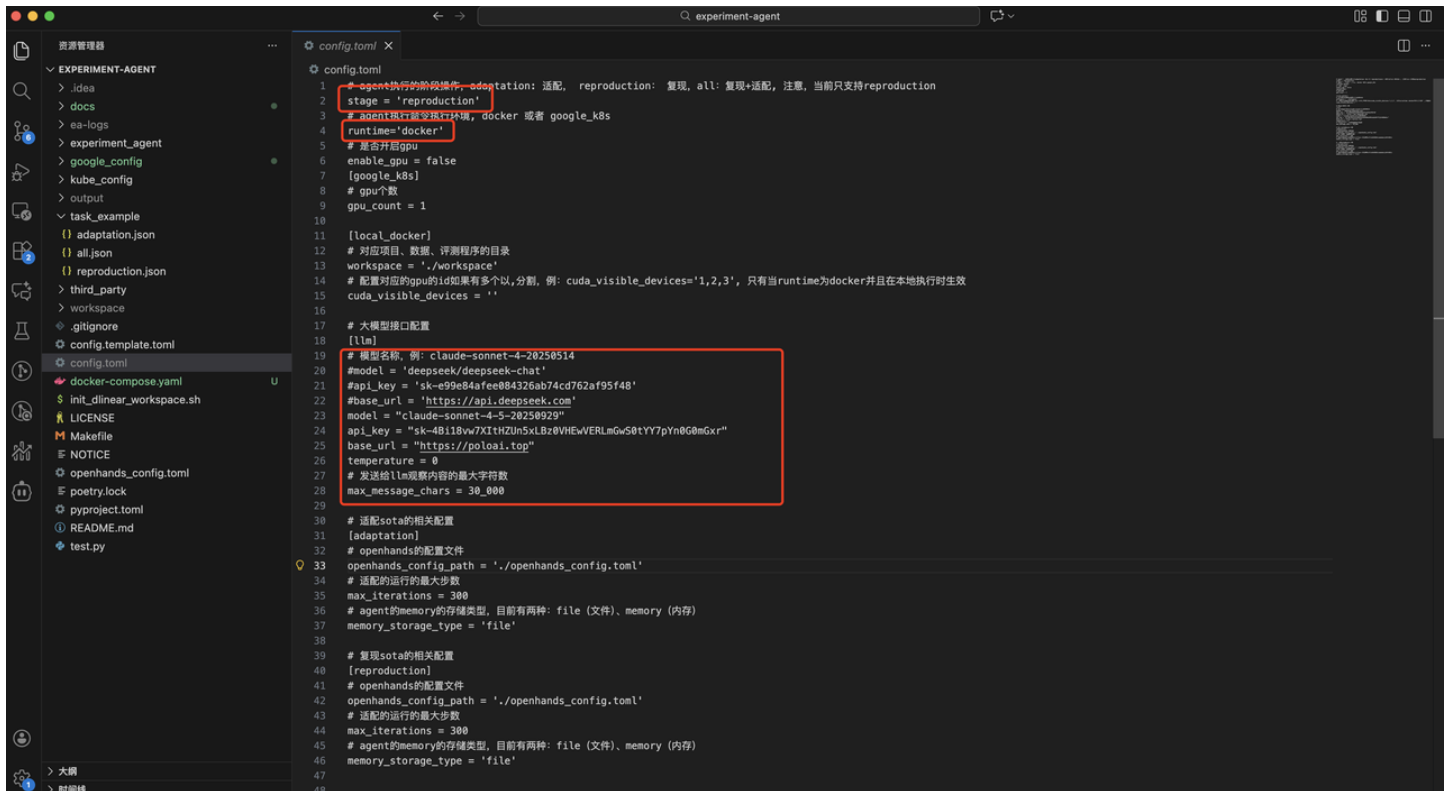
项目配置

检查项目根目录下config.toml文件配置

1.stage的值为reproduction

2.runtime的类型为docker

3.模型配置是否正确



任务配置检查

查看task_example/reproduction.json的值

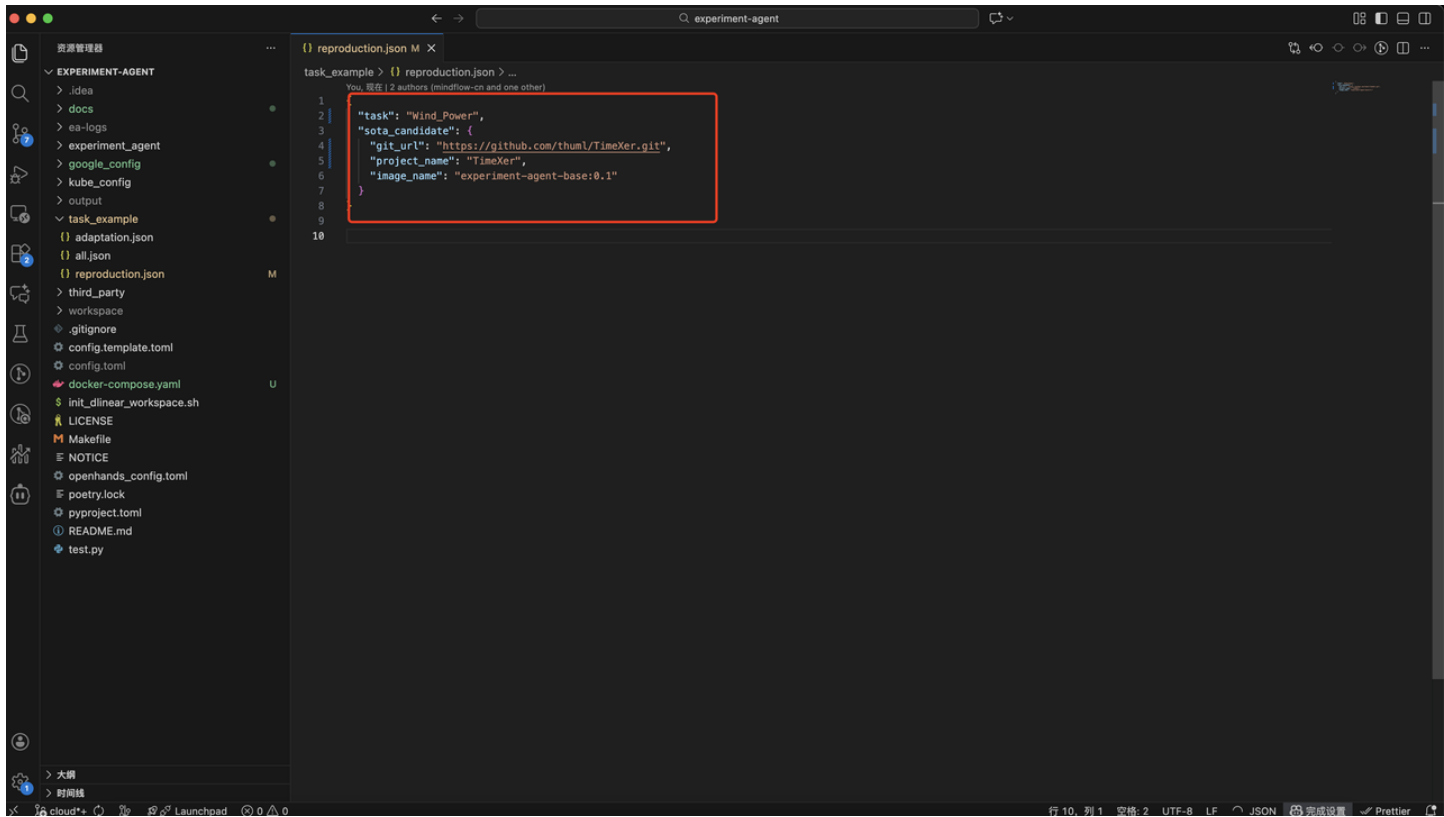
确认git_url是否为需要复现的项目地址:<https://github.com/thuml/TimeXer.git>

确认project_name是否为正确的项目名:TimeXer

确认image_name是否为构建的ea基础镜像

代码块

```
1 {
2   "task": "Wind_Power",
3   "sota_candidate": {
4     "git_url": "https://github.com/thuml/TimeXer.git",
5     "project_name": "TimeXer",
6     "image_name": "experiment-agent-base:0.1"
7   }
8 }
9
```

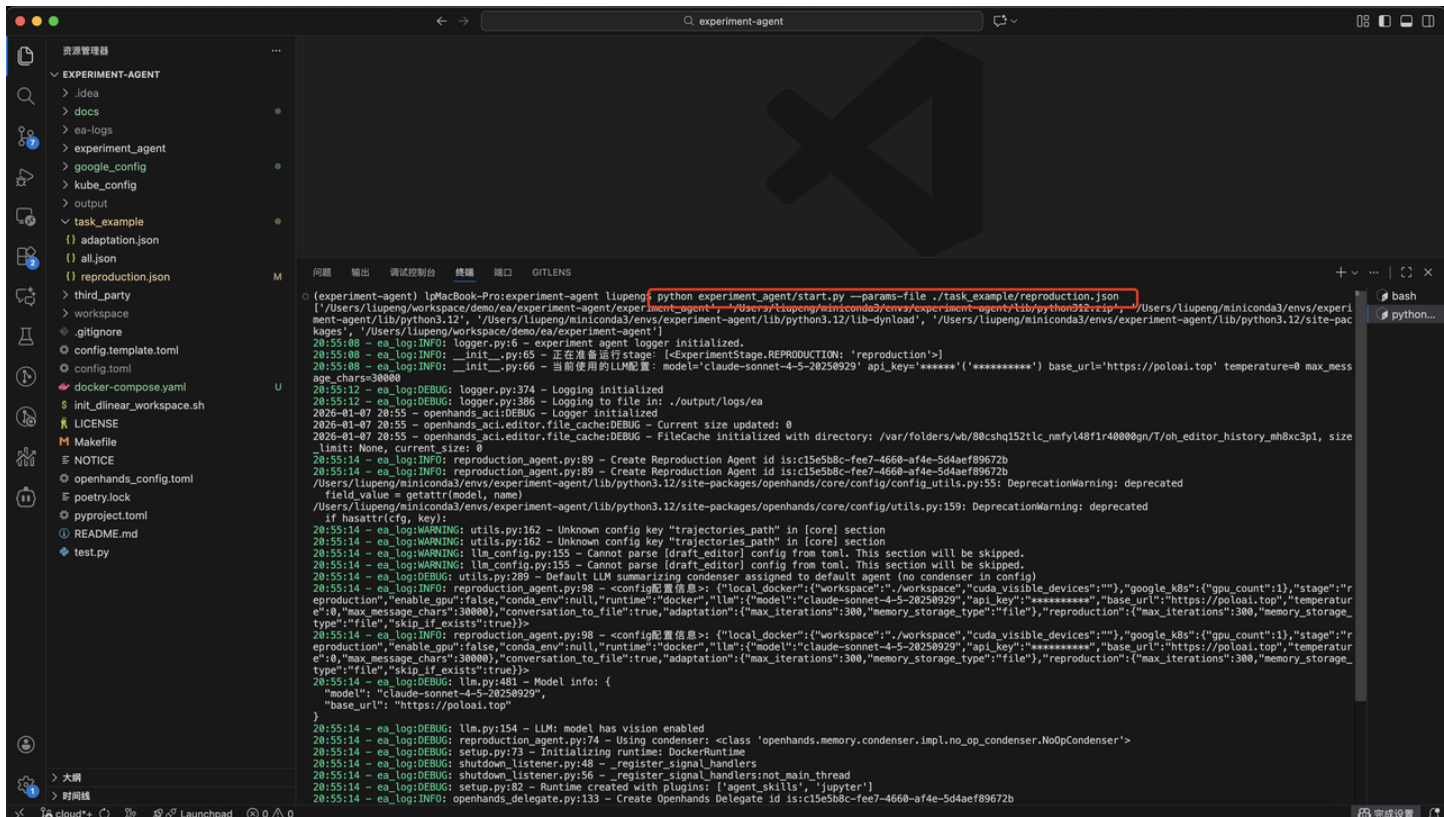


运行项目复现

运行复现命令

代码块

```
1 python experiment_agent/start.py --params-file
./task_example/reproduction.json
```



日志查看进入训练

The screenshot shows a VS Code interface with a project structure on the left, a terminal window in the center, and a Docker container panel on the right.

File Explorer (Left):

- .idea
- .vscode
- launch.json
- docs
- ea-logs
- experiment_agent
 - __pycache__
 -
 - adaptation
 - framework
 - output
 - reproduction
 - third_party
 - workspace
 - __init__.py
 - google_config
 - kube_config
 - start.py
 - output
 - task_example
 - adaptation.json
 - all.json
 - reproduction.json
 - third_party
 - workspace
 - .downloads
 - actionformer_release
 - TimeXer
 - checkpoints
 - data_provider
 - dataset/EPF
 - BE.csv
 - DE.csv
 - FR.csv
 - NP.csv
 - PJM.csv
 - exp

- 大纲
- 时间线

Terminal (Center):

```
13:06:16 - ea_log:DEBUG: bash.py:713 -      iters: 2100, epoch: 6 | loss: 0.1917935
      speed: 0.05275/iter; left time: 490.63575
13:06:21 - ea_log:DEBUG: bash.py:713 -      iters: 2200, epoch: 6 | loss: 0.1394798
      speed: 0.04835/iter; left time: 444.53265
13:06:25 - ea_log:DEBUG: bash.py:713 - Epoch: 6 cost time: 108.49707078933716
13:06:37 - ea_log:DEBUG: bash.py:713 - Epoch: 6, Steps: 2282 | Train Loss: 0.2212705 Vali Loss: 0.0679727 Test Loss: 0.0972285
Validation loss decreased (0.073041 -> 0.067973). Saving model ...
Updating learning rate to 3.125e-06
13:06:42 - ea_log:DEBUG: bash.py:713 -      iters: 100, epoch: 7 | loss: 0.0808991
      speed: 0.20965/iter; left time: 1892.51845
13:06:46 - ea_log:DEBUG: bash.py:713 -      iters: 200, epoch: 7 | loss: 0.0615975
      speed: 0.04365/iter; left time: 389.29175
13:06:51 - ea_log:DEBUG: bash.py:713 -      iters: 300, epoch: 7 | loss: 0.0761801
      speed: 0.05155/iter; left time: 454.61295
13:06:56 - ea_log:DEBUG: bash.py:713 -      iters: 400, epoch: 7 | loss: 0.0702799
      speed: 0.04825/iter; left time: 421.10335
13:07:02 - ea_log:DEBUG: bash.py:713 -      iters: 500, epoch: 7 | loss: 0.2136348
      speed: 0.05115/iter; left time: 441.27645
13:07:06 - ea_log:DEBUG: bash.py:713 -      iters: 600, epoch: 7 | loss: 0.0953770
      speed: 0.04915/iter; left time: 418.55925
13:07:11 - ea_log:DEBUG: bash.py:713 -      iters: 700, epoch: 7 | loss: 0.0937222
      speed: 0.04645/iter; left time: 391.34285
13:07:15 - ea_log:DEBUG: bash.py:713 -      iters: 800, epoch: 7 | loss: 0.1583427
      speed: 0.04415/iter; left time: 367.58095
13:07:20 - ea_log:DEBUG: bash.py:713 -      iters: 900, epoch: 7 | loss: 0.0653962
      speed: 0.04355/iter; left time: 357.90845
13:07:25 - ea_log:DEBUG: bash.py:713 -      iters: 1000, epoch: 7 | loss: 0.0659561
      speed: 0.05145/iter; left time: 417.50535
13:07:29 - ea_log:DEBUG: bash.py:713 -      iters: 1100, epoch: 7 | loss: 0.3491291
      speed: 0.04905/iter; left time: 393.05355
13:07:35 - ea_log:DEBUG: bash.py:713 -      iters: 1200, epoch: 7 | loss: 0.0531943
      speed: 0.04855/iter; left time: 384.92375
13:07:39 - ea_log:DEBUG: bash.py:713 -      iters: 1300, epoch: 7 | loss: 0.3519562
      speed: 0.04345/iter; left time: 339.47615
13:07:43 - ea_log:DEBUG: bash.py:713 -      iters: 1400, epoch: 7 | loss: 0.0945644
      speed: 0.04085/iter; left time: 315.08025
13:07:48 - ea_log:DEBUG: bash.py:713 -      iters: 1500, epoch: 7 | loss: 0.2030404
      speed: 0.04655/iter; left time: 354.78005
13:07:52 - ea_log:DEBUG: bash.py:713 -      iters: 1600, epoch: 7 | loss: 0.0632653
      speed: 0.04335/iter; left time: 326.32365
```

Docker Panel (Right):

- docker
- Python...